

Multi-Recognizer Multi-Attacker Adversarial Training for Robust Face Recognition

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Abstract—Face recognition models are no longer judged only by clean verification: black-box attacks, common corruptions, and broader robustness benchmarks show that deployed recognizers can fail under several kinds of shift and adversarial pressure. At the same time, robust learning has already used model or attack ensembles to reduce overfitting to a single adversary, and face recognition has developed several strong margin-based recognizer objectives. This work studies the intersection of these ideas: after introducing attacker diversity, does adding recognizer-objective diversity make adversarial training more rounded for one deployed target recognizer? The method treats robustness as a training-time pressure problem: a single ArcFace/iResNet100 target is trained on WebFace4M while adversarial examples are generated by multiple attack families and, in the full setting, also judged by external ArcFace, CosFace, and CurricularFace recognizer heads. The experiments are organized as an ablation from clean training to recognizer-only pressure, single-attack training, multi-attack training, and finally the full recognizer-plus-attacker setting. This separates the benefit of attack diversity from the additional benefit of recognizer diversity: the central comparison holds PGD, BPFA, and DFANet fixed, then adds the external recognizer heads. Adding this recognizer pressure raises average robust accuracy from 52.32% to 56.55% and reduces average attack success from 44.32% to 39.45%, but verification accuracy falls from 98.33% to 98.21% and EER rises from 1.78% to 1.93%. Inference cost is unchanged because only the target recognizer is deployed. The result is a calibrated tradeoff: recognizer diversity improves measured digital robustness beyond attack diversity alone, but shifts the model along a robustness-verification-training-cost frontier.

Index Terms—face recognition, adversarial training, ensemble robustness, ArcFace, verification, biometric security

I. INTRODUCTION

Face recognition (FR) systems are normally evaluated through clean identification or verification accuracy, but deployment failures often come from distribution shift and deliberate adversaries. Black-box attacks have been demonstrated against FR systems [8], adversarial FR surveys catalog multiple attack and defense families [9], and OODFace benchmarks degradation under corruptions and appearance variation [12]. Robust FR therefore requires not only clean identity classification but also stable embeddings under perturbed or shifted probes. Existing robustness recipes give useful foundations, especially robust optimization [6] and ensemble adversarial training [7], but face recognition has two additional complications. First, its decision surface is embedding-centered rather than only closed-set classification. Second, different margin-

based recognizers and different face-oriented attacks can create different failure modes.

This motivates a training-time robustness objective for a single deployed recognizer. Attack diversity probes the target from different perturbation procedures, while recognizer diversity probes the geometry induced by the identity objective. The cost is higher training pressure, memory use, and wall-clock time. The benefit, if the ablation supports it, is that robustness is absorbed into the target rather than shifted into a deployment stack of separate recognizers, detectors, or attack-specific modules. RobustBench and MultiRobustBench motivate this caution by showing how robustness can be overestimated when evaluation is too narrow [10], [11]. This project applies the same multi-pressure logic inside FR training and tests whether the resulting target moves to a better robustness point.

This work asks whether recognizer diversity adds anything once attack diversity is already present. The intuition is that ArcFace, CosFace, and CurricularFace shape the embedding space through different margin geometries, so using them as training-time judges may expose failures that a single ArcFace target would not see alone. The same pressure could also disturb the clean pairwise geometry needed for verification. The contribution is a controlled training framework that separates clean training, recognizer pressure, attack pressure, attack diversity, and their combination.

II. BACKGROUND AND DESIGN RATIONALE

The proposed training design has three key elements: the deployed target recognizer, the attacks used to generate hard examples, and the external recognizer objectives used only during training. The rationale for each part is different. Attack diversity addresses the risk of overfitting to one threat model, recognizer diversity addresses the geometry of the face embedding space, and large-scale FR training determines how much of this can be made practical.

The attacker side of the design follows from prior work on adversarial robustness. Robust optimization and ensemble adversarial training provide the foundation [6], [7], while RobustBench and MultiRobustBench show at benchmark scale that single-attack evaluation can overestimate practical robustness [10], [11]. FR-specific work provides the corresponding domain evidence: decision-based black-box attacks have been shown against real recognition systems [8], surveys document multiple perceptible and imperceptible attack families [9], and

OODFace extends reliability evaluation to corruptions and appearance variation [12].

The recognizer side comes from the structure of modern FR itself. ArcFace, CosFace, and CurricularFace all use normalized embeddings and margin-based classification, but they bias training differently: ArcFace enforces an additive angular margin, CosFace uses a cosine-margin formulation, and CurricularFace changes the treatment of hard examples during training [1]–[3]. They form a practical recognizer set because they are strong objectives with different decision geometries on the same embedding space. The central design question is whether attacker diversity should be paired with recognizer-objective diversity when hardening one deployed target.

Large-scale FR also constrains what can be tested. WebFace260M and WebFace42M/WebFace4M motivate million-image identity training [4], while Partial FC makes the large-class margin head trainable by sampling classifier weights [5]. The present experiments therefore test ensemble pressure in a large-identity regime where classifier-head memory, attack generation, and auxiliary recognizer losses all compete for GPU memory.

III. METHOD

A. Training Objective

The defended target is an ArcFace model with an iResNet100 backbone and a 512-dimensional embedding. Training uses WebFace4M manifests with RetinaFace-style aligned 112×112 crops. The experiments use a 205,849-identity WebFace4M subset obtained with a fixed random seed and a minimum of two images per identity.

The loss is constructed to make the source of pressure explicit. It combines clean classification, adversarial classification, embedding-consistency, optional recognizer-ensemble pressure, and hard-sample pressure:

$$\mathcal{L} = \lambda_c \mathcal{L}_c + \lambda_a \mathcal{L}_a + \lambda_r \mathcal{L}_r + \lambda_s \mathcal{L}_{sim} + \lambda_h \mathcal{L}_{hard}. \quad (1)$$

Here $\mathcal{L}_c$ and $\mathcal{L}_a$ keep the target discriminative on clean and adversarial images, $\mathcal{L}_r$ asks external recognizer heads to judge the same representations, $\mathcal{L}_{sim}$ discourages excessive clean/adversarial embedding drift, and $\mathcal{L}_{hard}$ emphasizes mined hard pairs or hard samples. The recognizer ensemble is used only as a training-time stressor; inference uses the single target recognizer.

B. Recognizer and Attacker Ensembles

The recognizer ensemble contains ArcFace, CosFace, and CurricularFace heads, three strong and widely used margin-loss variants. The attacker suite contains PGD, BPFA, and DFANet. PGD is the white-box optimization anchor, BPFA adds transfer-oriented pressure, and DFANet adds FR-specific feature-level pressure. This suite does not cover every modern attack; it provides a controlled test of whether different attacker and recognizer pressures improve a single target.

Training uses a three-stage curriculum: clean warmup, shallow adversarial pressure, then full adversarial pressure. In

the main 24-epoch schedule, clean warmup lasts 3 epochs, shallow adversarial training lasts 9 epochs, and full adversarial training lasts 12 epochs. The shallow and full phases use the same attack families but not the same attack strength: shallow training uses reduced budgets and restart depth so that adversarial pressure is introduced before the strongest inner loops dominate optimization, while the final phase uses the full primary-attacker rotation. The full condition uses a 50% clean / 50% adversarial batch split with one primary attacker family active per adversarial batch. Adversarial intent is also balanced between impersonation and dodging, matching the two main failure modes of face-recognition systems.

C. Practical Training Pipeline

Each iteration is arranged around a deployable target branch and training-only pressure branches. Clean images always update the ArcFace/iResNet100 target through the margin objective. During adversarial stages, a fixed portion of the batch is copied into an attack branch, where one attacker family is chosen in round-robin order and optimized against the current target/surrogate state. The online attacks use random starts and restart selection where configured, so the adversarial branch is not only a deterministic augmentation path. The generated adversarial images then re-enter the same target branch for adversarial classification and clean/adversarial embedding consistency. This keeps the model being optimized identical to the model being evaluated.

Recognizer pressure is attached after the target representation has been formed rather than by deploying multiple recognizers. The ArcFace, CosFace, and CurricularFace recognizer objectives evaluate the same clean or adversarial samples and contribute auxiliary losses during training, with weighted recognizer terms to prevent a single head from dominating. The target Partial FC rate is 0.30, while auxiliary recognizer heads use a lower sampling rate of 0.12 to avoid turning the recognizer ensemble into a memory bottleneck. The attacker ensemble is similarly implemented as scheduled pressure rather than as one combined adversarial batch: PGD, BPFA, and DFANet are rotated across adversarial batches, while attack generation itself is chunked to fit the large iResNet100 target and Partial FC classifier into 24 GB GPUs. The hard sample term is also staged, so early epochs first learn identity structure before harder mined pairs receive substantial weight. The final trained model remains one target recognizer; the ensemble exists only in the training graph.

D. Comparison Sequence

The comparison sequence isolates one source of pressure at a time. *Baseline-clean* sets the non-robust recognition reference. *Recognizer-ensemble-only* tests whether recognizer pressure alone changes clean embedding quality. *Single-attack-only* tests the risk of narrow adversarial training. *Attack-ensemble-only* tests the conventional multiattack improvement. *Full-ensemble* then adds recognizer diversity on top of the same attacker suite, making it the central comparison for the main claim.

E. Threat Model and Evaluation Boundary

The evaluated attacks are digital image-space attacks, not physical or presentation attacks. PGD is treated as a white-box optimization attack, BPFA as transfer-oriented pressure, and DFANet as an FR feature-level attack. During training, attacks are generated against the current target/surrogate state, so the setup is not defense by obscurity. Robust classification is identification-style: it asks whether the closed-set identity prediction survives perturbation. Pair verification instead asks whether embeddings still separate same-identity from different-identity pairs.

The evaluation is not adaptive. It does not test an adversary that optimizes through the complete training-time recognizer ensemble, tunes attacks after observing all defense hyperparameters, or performs query-based score/decision attacks against a deployed system. The claim is therefore restricted to the PGD/BPFA/DFANet digital suite and the ablation question of whether recognizer-objective pressure adds robustness once attacker diversity is already present.

IV. EXPERIMENTAL SETUP AND METRICS

All main experiments use ArcFace/iResNet100 on WebFace4M with 4,015,275 training images and 219,826 validation images over 205,849 identities. Partial FC uses a 0.30 negative sampling rate to keep large-class training feasible on 8 RTX 4090D GPUs. Main per-GPU batch settings are clean warmup 128, shallow adversarial 48, full adversarial 32, and attack chunk size 32. Mixed precision, channels-last tensors, distributed training, and manifest-backed dataloading are used.

Metrics are split by the question they answer. Closed-set validation measures identity classification; robust validation measures adversarial accuracy, ASR, clean-to-robust drop, and per-attack behavior; verification measures pairwise accuracy, EER, ROC-AUC, TAR at low FAR, and cosine separation. Verification is in-domain: pairs are sampled from the WebFace4M training manifest and validation identities overlap training identities, so it is an embedding-geometry analysis rather than an external generalization claim. It deduplicates pair images, extracts embeddings in distributed GPU batches, scores cosine similarities in chunks, and computes ROC statistics from sorted scores. Training cost is reported with wall time, GPU-hours, throughput, attack-generation share, peak CUDA memory, and active per-GPU batch sizes; inference cost is unchanged because the auxiliary recognizers and attackers are discarded after training.

Several implementation choices were made specifically to keep the full setting trainable rather than merely conceptually defined. The recognizer pressure is stage-gated so the clean warmup remains fast; auxiliary recognizer heads use a lower Partial FC sampling rate than the target; attack batches are chunked; verification is batched and distributed; and the full-adversarial batch size is kept at 32 after larger batch trials increased memory pressure without improving stable throughput. Training attackers and evaluation attackers are kept separate in configuration: clean-only and recognizer-only conditions receive no adversarial training pressure, but the

final robustness table evaluates all conditions against the same PGD/BPFA/DFANet suite.

The evaluation suite is narrower than the future threat model. Stronger AutoAttack/APGD-style checks, TransMix-style transfer attacks, AdvFaceGAN, Adv-Makeup, Greedy-DiM morphing, diffusion-based attacks, and physical or PAD-aware stress tests are not counted as evidence here because they require faithful separate implementations or generated-image protocols [16]–[18].

V. RESULTS

A. Overview

The results support a conditional claim. Recognizer pressure alone preserves clean in-domain verification but does not make the target robust. Narrow PGD-only pressure is unstable and damages both classification and verification. The full method becomes useful after the attacker side is already diversified: ArcFace/CosFace/CurricularFace pressure improves robustness but moves the embedding space away from the clean-verification optimum.

B. Robustness

TABLE I
CLOSED-SET CLEAN AND ADVERSARIAL ROBUSTNESS RESULTS. ROBUST AVERAGE IS OVER PGD/BPFA/DFANET UNLESS MARKED OTHERWISE.

Condition	Clean Acc.	Robust Acc.	Drop	ASR
Baseline-clean	99.23	0.21	99.02	99.79
Recognizer-only	99.23	0.23	99.01	99.77
Single-attack	79.69	34.51	45.18	55.09
Attack-ensemble	93.98	52.32	41.65	44.32
Full-ensemble	93.40	56.55	36.85	39.45

All numbers are percentages. Robust accuracy and ASR are evaluated over PGD/BPFA/DFANet.

Table I answers the primary robustness question by holding the attacker suite fixed. Moving from attack-ensemble-only to full-ensemble raises robust accuracy from 52.32% to 56.55%, lowers ASR from 44.32% to 39.45%, and reduces the clean-to-robust drop from 41.65 to 36.85 points. The controlled difference between two rows that share PGD, BPFA, and DFANet is the empirical signal for recognizer-objective diversity.

The added full-suite evaluations sharpen the lower end of the comparison sequence. Clean and recognizer-only models are almost completely vulnerable under the three-attack suite, with robust accuracy near 0.2000% and ASR near 99.80%. The recognizer-only condition therefore improves verification slightly but does not create adversarial robustness by itself. Single-attack training is a partial case: the three-attack average is 34.51%, with BPFA at 45.71% but PGD and DFANet both near 28.90%. That pattern suggests PGD-only training does not transfer evenly across attack families.

The full-versus-attack-ensemble per-attack structure is more informative than the average. PGD improves from 45.45% to 52.55%, BPFA from 66.09% to 69.56%, and DFANet from 45.43% to 47.55%. The largest gain on PGD suggests

that recognizer pressure helps the target resist direct margin-optimization rather than transfer-style noise. BPFA remains the easiest attack after training, consistent with transfer pressure being partly absorbed by the diversified attacker ensemble. DFANet improves least, which is consistent with feature-level FR attacks aligning more closely with the embedding objective being optimized.

The clean-accuracy movement complicates the robustness gain. Full-ensemble clean accuracy is slightly lower than attack-ensemble-only (93.40% versus 93.98%), so the recognizer heads do not behave like clean regularizers. They appear to reshape the representation toward perturbation stability. The single-attack row is more severe: PGD-only training reaches 79.69% clean accuracy and 34.51% average robust accuracy, consistent with over-specialized adversarial pressure degrading the margin classifier while leaving substantial cross-attack vulnerability.

C. Verification

TABLE II

IN-DOMAIN WEBFACE4M PAIR-VERIFICATION METRICS. TAR IS REPORTED AT LOW FAR OPERATING POINTS, FOLLOWING STANDARD FR REPORTING PRACTICE.

Condition	Verif. Acc.	EER	ROC-AUC	TAR@1e-4	TAR@1e-5
Baseline-clean	99.96	0.0470	99.999	99.86	99.65
Recognizer-only	99.96	0.0410	99.999	99.92	99.73
Single-attack	89.74	12.82	92.01	0.37	0.11
Attack-ensemble	98.33	1.78	99.75	85.02	73.38
Full-ensemble	98.21	1.93	99.73	84.81	74.16

All numbers are percentages. Lower EER is better; higher values are better for all other columns.

Table II is an embedding-geometry analysis rather than an external generalization benchmark. The verification bundle contains 409,196 WebFace4M pairs from the training manifest, and validation identities overlap training identities. Near-ceiling clean verification is expected; the relevant signal is the relative degradation pattern across rows, especially at low-FAR operating points and in cosine separation.

The clean and recognizer-only rows provide the control. Both retain about 99.96% verification accuracy, near-perfect ROC-AUC, and EER below 0.0500%. The recognizer-only condition slightly improves EER and low-FAR TAR, so the external margin heads are not inherently harmful to the embedding space. Degradation appears when margin diversity is coupled to adversarial image pressure, where the target must satisfy both clean identity separation and perturbed-feature stability.

The single-attack row exposes the most severe geometric failure: 89.74% verification accuracy, 12.82% EER, 92.01% ROC-AUC, and only 0.37% TAR@1e-4. That low-FAR collapse indicates a score-distribution shift, not only a loss of class accuracy. For FR, this makes closed-set robust accuracy insufficient on its own.

Attack-ensemble training restores much of this structure while providing robustness: 98.33% verification accuracy, 1.78% EER, and 99.75% ROC-AUC. The full ensemble then

improves robust accuracy at a small but coherent verification cost: 98.21% accuracy, 1.93% EER, and TAR@1e-4 of 84.81%. The mean negative cosine increases from 0.0440 to 0.0640, indicating weaker negative-pair separation. The full model is harder to attack, but its adversarial invariance appears to compress part of the clean verification geometry.

The full method is a movement along a robustness–verification frontier. For a deployment where adversarial probes are plausible, the 4.23-point robust accuracy gain and 4.87-point ASR reduction may justify the roughly 0.12-point verification-accuracy loss and 0.15-point EER increase relative to attack-ensemble-only. For a deployment dominated by clean verification, the clean or recognizer-only rows remain preferable. This distinction is the study’s main interpretive result: multi-recognizer multi-attacker training is robustness pressure, not a universal replacement for clean FR training.

D. Training Cost

TABLE III

TRAINING COST MEASURED FROM COMPLETED 24-EPOCH HISTORIES ON 8 GPUS. WALL TIME INCLUDES TRAINING, VALIDATION, VERIFICATION, AND MODEL-SAVING OVERHEAD. BATCH LISTS THE DOMINANT ACTIVE PER-GPU BATCH SIZE.

Condition	Batch	Wall h	GPU h	Img/s	Attack Share	Peak GB
Baseline-clean	128	12.70	101.9	262.8	0.0000	20.30
Recognizer-only	128	26.10	208.9	128.1	0.0000	20.60
Single-attack	48/32	37.60	300.9	89.00	55.70	20.30
Attack-ensemble	48/32	75.50	604.2	44.30	74.90	20.30
Full-ensemble	48/32	110.1	880.7	32.70	67.70	22.60

Img/s is end-to-end average throughput over the completed training history. Attack share is attack generation as a fraction of measured train compute.

Table III reports the cost of the strongest implemented pressure point under a memory-capped batch schedule. The clean baseline takes 12.70 wall-clock hours. Recognizer-only training roughly doubles the cost to 26.10 hours because the auxiliary recognizer heads require additional backbone/head passes even without attacks. Single-attack training takes 37.60 hours, attack-ensemble training takes 75.50 hours, and the full condition reaches 110.1 hours, or 880.7 GPU-hours on the 8-GPU setup. Wall time mixes algorithmic cost with a memory-cap artifact: clean and recognizer-only conditions train at batch 128, while adversarial phases are capped at 48 and 32. The baseline therefore uses up to four times the full-adversarial batch size for a significant amount of training time.

The dominant cost is the adversarial inner loop. Attack-ensemble-only spends 74.90% of measured train compute on attack generation, and the full ensemble still spends 67.70% there despite the added recognizer losses. Each attacked batch performs repeated target/surrogate forward-backward passes before the ordinary parameter update; the full condition also evaluates recognizer heads and samples large-class margin weights. Peak memory reaches 22.60 GB on 24 GB cards, so the method is limited both by repeated computation and by reduced batch size. The bottleneck is therefore not the dataloader but repeated adversarial optimization under a large-class FR head.

The full ensemble gains 4.23 robust-accuracy points over attack-ensemble-only at the cost of 34.60 additional wall-clock hours and 276.50 GPU-hours. The marginal robustness gain is therefore expensive. Existing acceleration work suggests several practical directions: recycling gradients or adversarial states as in Free Adversarial Training [13], reducing repeated full backpropagation as in YOPO [14], or using cheaper single-step adversaries with safeguards against catastrophic overfitting [15]. For this FR setting, the closest direct ablations are lower-frequency recognizer pressure, cached or replayed adversarial seeds, cheaper early-epoch attack curricula, shared surrogate forwards across attacks, and selective full-depth attack refreshes. These target the measured bottleneck without departing from the multi-recognizer multi-attacker design objective.

VI. DISCUSSION AND LIMITATIONS

The work is strongest as an ablation-backed framework result. It does not show state-of-the-art FR, that every ensemble member is beneficial, or that the selected attacks are sufficient for deployment. It shows a narrower result: when a target already receives multiattack pressure, adding recognizer-objective diversity improves measured robustness under the same attacker suite. The single-attack result also suggests that narrow PGD-only training is weaker than broader attack-family training.

Several limitations remain. First, the attacker suite is explicit but not exhaustive. Second, verification and closed-set robust classification answer different questions, and both are needed for FR robustness. Third, the verification protocol is in-domain; external unseen-identity benchmarks such as LFW, CFP-FP, AgeDB, or IJB-style evaluations are needed before deployment claims. Fourth, the recognizer ensemble is training-time pressure, not a deployment ensemble. Fifth, training cost affects the design itself: memory-limited batch sizes, repeated attack forwards, and large-class margin heads make the full condition much slower than the clean baseline. Finally, the work does not yet establish robustness to adaptive ensemble attacks, modern semantic or generative attacks, morphing attacks, corruptions, OOD appearance variation, or physical/PAD-aware attacks.

VII. CONCLUSION

This study implements a face-recognition robustness framework that trains one ArcFace/iResNet100 target under explicit recognizer and attacker ensemble pressure. The result is an ablation-backed tradeoff rather than a simple leaderboard win. Recognizer-objective diversity adds measured adversarial robustness beyond attacker diversity, but the gain comes with lower clean-verification performance and a large training-cost increase. The method is therefore best understood as a practical training framework for FR hardening: useful when digital adversarial robustness is part of the operating requirement, less attractive when clean verification is the only priority, and still bounded by the attacks and in-domain verification protocol evaluated here.

REFERENCES

- [1] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4690–4699.
- [2] H. Wang, Y. Wang, Z. Zhou, X. Ji, D. Gong, J. Zhou, Z. Li, and W. Liu, "CosFace: Large margin cosine loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018.
- [3] Y. Huang, Y. Wang, Y. Tai, X. Liu, P. Shen, S. Li, J. Li, and F. Huang, "CurricularFace: Adaptive curriculum learning loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020.
- [4] Z. Zhu *et al.*, "WebFace260M: A benchmark unveiling the power of million-scale deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021.
- [5] X. An *et al.*, "Partial FC: Training 10 million identities on a single machine," arXiv:2010.05222, 2020.
- [6] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [7] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "Ensemble adversarial training: Attacks and defenses," in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [8] Y. Dong, H. Su, B. Wu, Z. Li, W. Liu, T. Zhang, and J. Zhu, "Efficient decision-based black-box adversarial attacks on face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7714–7722.
- [9] F. Vakhshiteh, A. Nickabadi, and R. Ramachandra, "Adversarial attacks against face recognition: A comprehensive study," *IEEE Access*, vol. 9, pp. 92735–92756, 2021.
- [10] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, "RobustBench: A standardized adversarial robustness benchmark," in *Adv. Neural Inf. Process. Syst.*, 2021.
- [11] S. Dai, S. Mahloujifar, C. Xiang, V. Sehwag, P.-Y. Chen, and P. Mittal, "MultiRobustBench: Benchmarking robustness against multiple attacks," in *Proc. Int. Conf. Mach. Learn.*, 2023.
- [12] C. Kang, Y. Chen, S. Ruan, S. Zhao, R. Zhang, J. Wang, S. Fu, and X. Wei, "OODFace: Benchmarking robustness of face recognition under common corruptions and appearance variations," arXiv:2412.02479, 2024.
- [13] A. Shafahi, M. Najibi, M. A. Ghiasi, Z. Xu, J. Dickerson, C. Studer, L. S. Davis, G. Taylor, and T. Goldstein, "Adversarial training for free!" in *Adv. Neural Inf. Process. Syst.*, 2019.
- [14] D. Zhang, T. Zhang, Y. Lu, Z. Zhu, and B. Dong, "You only propagate once: Accelerating adversarial training via maximal principle," in *Adv. Neural Inf. Process. Syst.*, 2019.
- [15] E. Wong, L. Rice, and J. Z. Kolter, "Fast is better than free: Revisiting adversarial training," in *Proc. Int. Conf. Learn. Represent.*, 2020.
- [16] B. Yin, W. Wang, T. Yao, J. Guo, Z. Kong, S. Ding, J. Li, and C. Liu, "Adv-Makeup: A new imperceptible and transferable attack on face recognition," in *Proc. Int. Joint Conf. Artif. Intell.*, 2021, pp. 1252–1258.
- [17] Z. W. Blasingame and C. Liu, "Greedy-DiM: Greedy algorithms for unreasonably effective face morphs," in *Proc. IEEE Int. Joint Conf. Biometrics*, 2024.
- [18] H. Huang, Y. Yang, and Y. Wang, "AdvFaceGAN: A face dual-identity impersonation attack method based on generative adversarial networks," *PeerJ Computer Science*, vol. 11, e2904, 2025.